Alex Morgan

Curriculum Vitae

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Education

New York University — B.S. in Computer Science Sep 2020 – May 2024

New York, NY

- Minor in Mathematics | GPA: 3.88 / 4.0 | Dean's Honor List
- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Systems Organization, Database Systems, Computer Networks, Distributed Systems, Machine Learning

Research & Professional Experience

Software Engineering Intern Jun 2024 – Aug 2024

Datadog · New York, NY

- Engineered high-throughput event streaming pipelines using Go and Apache Kafka, processing 50M+ telemetry metrics daily with sub-second latency.
- Designed and deployed containerized RESTful microservices with Docker and Kubernetes, reducing API response times by 35%.
- Implemented automated end-to-end integration test suites in CI/CD pipelines, increasing branch test coverage from 78% to 93%.

Undergraduate Research Assistant Sep 2023 – May 2024

NYU Courant Machine Learning and Systems Lab · New York, NY

- Researched memory-efficient attention mechanisms and distributed training optimizations for transformer models in PyTorch.
- Profiled GPU kernel execution with NVIDIA Nsight and CUDA to identify memory bandwidth bottlenecks during multi-GPU inference.
- Developed scalable data preprocessing pipelines in Python and Ray to clean and tokenize 2TB+ multimodal benchmark datasets.

Software Engineering Intern Jun 2023 – Aug 2023

MongoDB · New York, NY

- Built full-stack telemetry dashboards using TypeScript, React, and Node.js for real-time database cluster health monitoring.
- Optimized aggregation pipeline queries for distributed document collections, improving query execution latency by 40%.
- Collaborated across an agile team of 6 engineers using Git, code reviews, and bi-weekly sprint planning.

Undergraduate Teaching Assistant Jan 2023 – May 2024

NYU Department of Computer Science · New York, NY

- Led weekly recitation sessions and lab hours for 120+ students in Data Structures and Algorithms.
- Held 1-on-1 office hours to provide debugging guidance and algorithmic problem-solving strategies in Java and Python.
- Graded weekly programming projects, written assignments, and exams with comprehensive feedback.

Publications & Preprints

- Chen, H., **Morgan, A.**, LeCun, Y., Bruna, J. (2024). *Adaptive Attention Caching for Low-Latency Transformer Inference.* preprint, arXiv:2404.12345. [Code].
- **Morgan, A.**, Zhang, L., Shasha, D. (2023). *Fault-Tolerant State Replication in Heterogeneous Edge Environments.* Proceedings Paper, IEEE International Conference on Distributed Computing Systems (ICDCS Workshops).

Presentations & Posters

- **Morgan, A.**, Chen, H., Bruna, J. (Apr 2024). *Optimizing Memory Bandwidth in Multi-GPU Transformer Training*. Oral Presentation, NYU Courant Undergraduate Research Showcase, New York, NY.
- **Morgan, A.**, Zhang, L. (Oct 2023). *Consensus Protocol Performance Under High Network Latency*. Poster Presentation, ACM Richard Tapia Celebration of Diversity in Computing, Dallas, TX.

Projects

RaftKV — Distributed Fault-Tolerant Key-Value Store Jan 2024 – Apr 2024

- Engineered a distributed key-value store in Go implementing the Raft consensus protocol for high availability and strong consistency.
- Supported leader election, log replication, dynamic cluster reconfiguration, and snapshot compaction.
- Benchmarked performance under simulated network partitions, sustaining 15,000+ requests/sec across 5-node clusters.

CloudNet — Containerized Microservices Platform Sep 2023 – Dec 2023

- Architected a scalable microservices web application using Python (FastAPI), React, PostgreSQL, and Redis caching.
- Automated continuous integration and deployment with GitHub Actions and Helm charts to a Kubernetes cluster.
- Integrated Prometheus metrics and Grafana dashboards for real-time monitoring of service latency and resource usage.

NeuralCode — Semantic Code Search & Summarization Feb 2023 – May 2023

- Fine-tuned a CodeBERT transformer model using PyTorch and Hugging Face to generate natural language docstrings from Python source code.
- Built an interactive web interface with Next.js and Milvus vector database for sub-50ms natural language semantic code search.

Skills

Languages:	Python, Go, C/C++, Java, TypeScript, JavaScript, SQL, Rust, Bash
Frameworks & Libraries:	React, Node.js, FastAPI, Next.js, PyTorch, Express, GraphQL, gRPC
Systems & Cloud:	Docker, Kubernetes, Linux / Unix, AWS (EC2, S3, Lambda), PostgreSQL, Redis, Apache Kafka, Git, CI/CD (GitHub Actions)
Core Concepts:	Distributed Systems, Algorithms & Data Structures, Object-Oriented Design, RESTful APIs, Microservices, Machine Learning

Leadership & Service

President & Workshop Lead May 2023 – May 2024

ACM Student Chapter, New York University · New York, NY

- Organized 15+ technical workshops on systems programming, technical interview preparation, and cloud computing for 300+ members.
- Directed HackNYU hackathon logistics, securing $20,000 in industry sponsorships and coordinating 400+ participants.

Peer Mentor Sep 2022 – May 2024

Department of Computer Science, New York University · New York, NY

- Mentored 20+ first-year computer science undergraduates on course planning, study strategies, and research opportunities.
- Facilitated bi-weekly collaborative study circles for foundational computer science and discrete mathematics courses.

Honors & Awards

- **Dean's Undergraduate Research Fund (DURF) Grant**, College of Arts & Science, New York University Spring 2024
- **Courant Academic Excellence in Computer Science Award**, Courant Institute of Mathematical Sciences, New York University May 2024
- **1st Place Winner — Systems & Infrastructure Track**, HackNYU Annual Hackathon Feb 2023
- **Dean's Honors List**, New York University Fall 2020 – Spring 2024